

Summary

This video elucidates the core principles of discrete probability distributions, focusing on the **Bernoulli** and **Binomial** models.

- The **Bernoulli distribution** is the fundamental building block.
- A Bernoulli trial consists of a single experiment with two mutually exclusive outcomes:
 - Success (p)
 - Failure (q)
- The **Binomial distribution** extends the Bernoulli model to a fixed number of trials (n).
- The derivation involves combinatorial logic and Pascal's Triangle.
- Key statistical parameters include Mean, Variance, and Standard Deviation.

I. Bernoulli Distribution: The Fundamental Trial

The Bernoulli distribution is the simplest discrete probability distribution.

Core Characteristics

- **Discrete Nature:** Random variable X takes integer values (0 or 1).
- **Two Outcomes:**
 - Success = 1
 - Failure = 0
- **Independence:** Outcomes do not influence each other.
- **Single Trial:** $n = 1$.

Mathematical Representation

If p = probability of success

q = probability of failure

$$q = 1 - p$$

$$p + q = 1$$

Probability Mass Function (PMF)

$$P(X = x) = p^x (1 - p)^{(1 - x)}$$

where $x = 0$ or 1 .

Statistical Parameters for Bernoulli

Parameter	Formula
Mean (Expected Value)	p
Variance	$p q$ or $p(1 - p)$
Standard Deviation	$\sqrt{p q}$

II. Binomial Distribution: Scaling the Model

The Binomial distribution applies when multiple Bernoulli trials are performed.

If a Bernoulli trial is a single building block, the Binomial distribution is constructed from multiple such trials.

Requirements for a Binomial Distribution

1. Fixed number of trials (n)
2. Constant probability of success (p)
3. Independence of trials
4. Allows r successes ($r \geq 0$)

Role of Combinations

To determine probability of r successes in n trials:

Number of ways = nCr

This can be obtained using Pascal's Triangle.

Binomial Probability Mass Function (PMF)

$$P(X = r) = nCr \cdot p^r \cdot q^{(n - r)}$$

Parameters of Binomial Distribution

Defined by two parameters: n and p .

If n and p are known, the distribution is fully defined.

Parameter	Formula
Mean (Expected Value)	np
Variance	npq
Standard Deviation	$\sqrt{npq}$

III. Comparative Summary

Feature	Bernoulli Distribution	Binomial Distribution
Number of Trials	Exactly 1 ($n = 1$)	Fixed $n > 1$
Possible Outcomes	0 or 1	$0, 1, 2, \dots, n$
Parameters	1 (p)	2 (n, p)
Mean	p	$n p$
Variance	$p q$	$n p q$

IV. Application Examples and Problem Logic

1. Multi-Colored Die Problem

Scenario:

- A die has opposite faces colored identically (Red, Blue, Green).
- Each color appears twice.
- Die thrown 3 times ($n = 3$).
- Success = Red.

Probability:

$$P(\text{Red}) = 2/6 = 1/3$$

$$P(\text{Not Red}) = 4/6 = 2/3$$

To find probability of at least twice:

$$P(X = 2) + P(X = 3)$$

Use Binomial formula.

2. Five-Coin Sum Problem

Scenario:

- Five coins marked with 2 and 3.
- Find probability that total sum = 12.

Logic:

Only combination possible:

3 twos and 2 threes

$$3 \times 2 + 2 \times 3 = 12$$

If “2” is success:

$r = 3$ successes

$n = 5$ trials

Apply Binomial formula.

3. Manufacturing Quality Control

Scenario:

- Probability of defective bolt $p = 0.1$
- Total bolts $n = 900$

Mean:

$$n p = 900 \times 0.1 = 90$$

Variance:

$$n p q = 900 \times 0.1 \times 0.9 = 81$$

Standard Deviation:

$$\sqrt{81} = 9$$

V. Key Definitions

- **Discrete Distribution:** Values that can be counted (0,1,2,3,...).
- **Trial Interpretation:** Each independent attempt is a Bernoulli trial.
- **Parameter Dependency:** Binomial distribution depends on n and p .
- **Success and Failure:** Every trial has exactly two outcomes.

